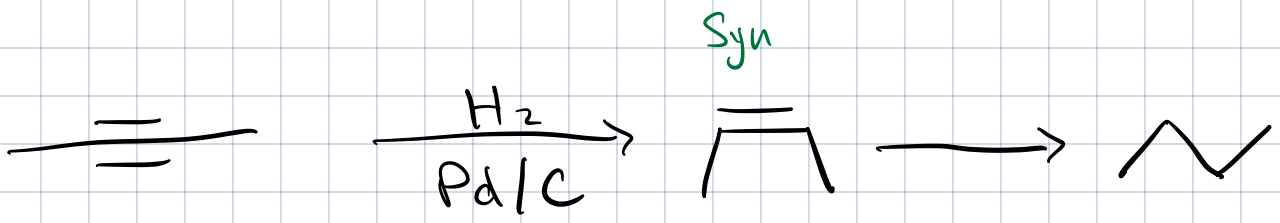
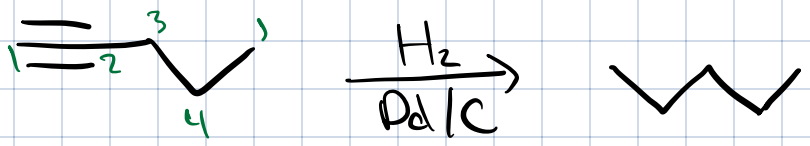
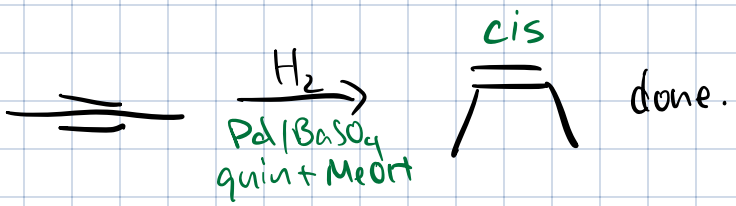
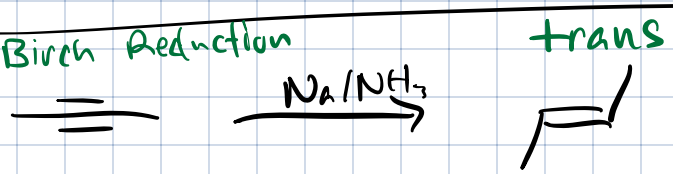




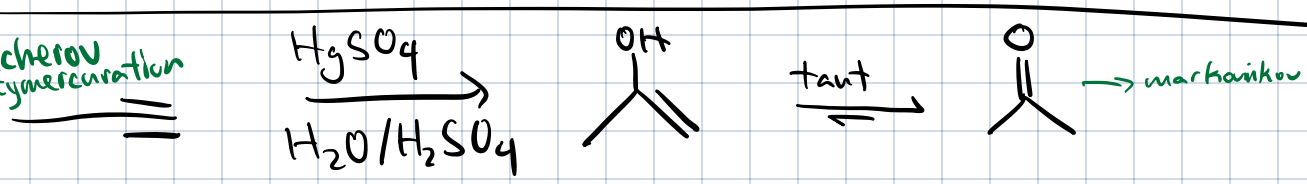
Lindlar (poison) catalyst (syn hydrogenation kyne → kene)



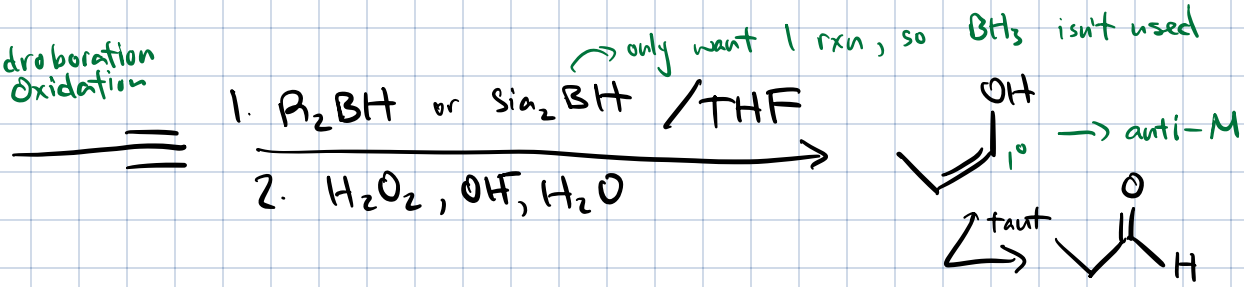
Birch Reduction

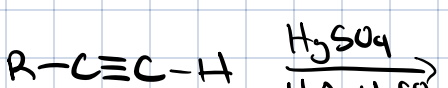


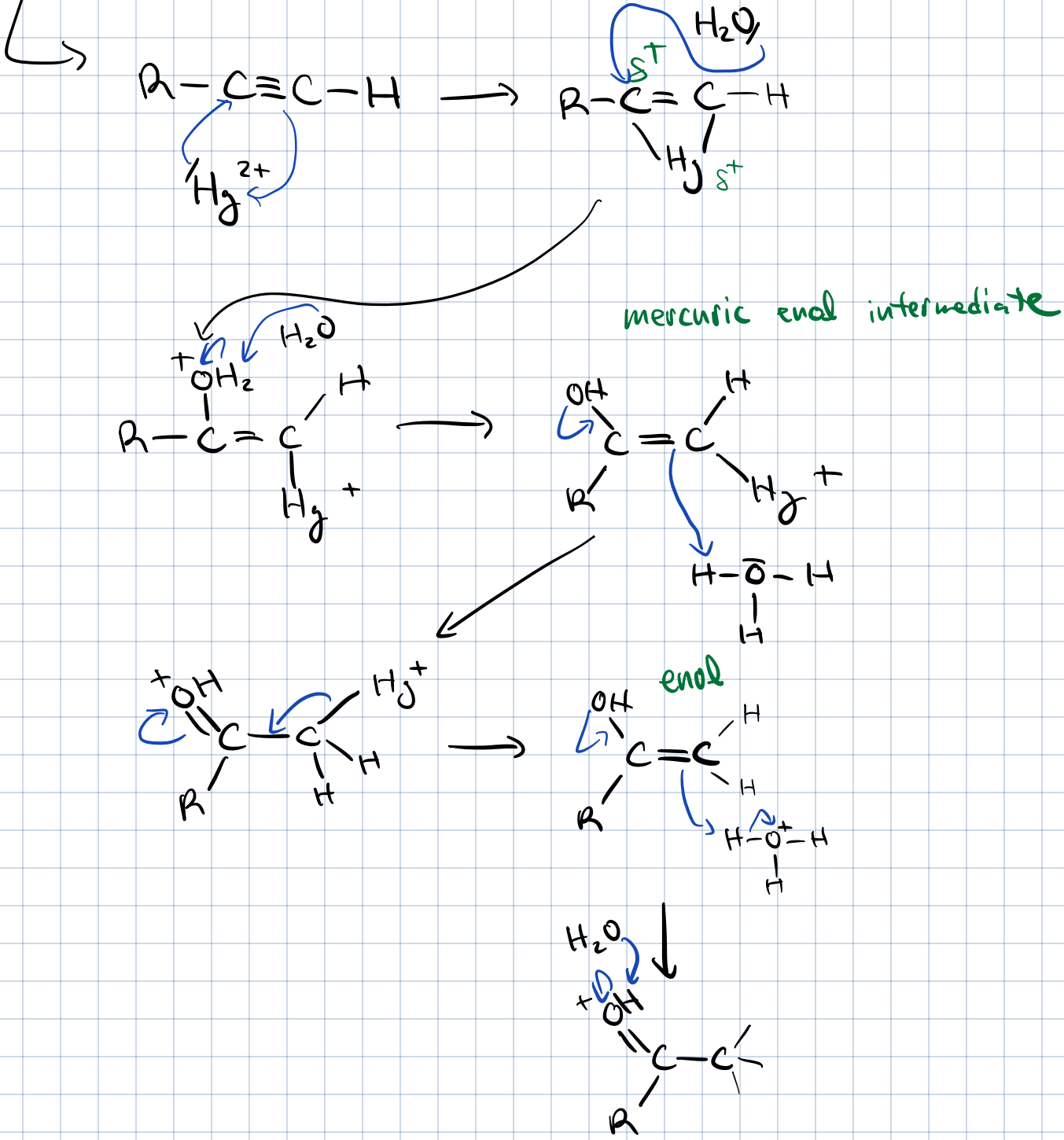
Kucherov oxymercuration

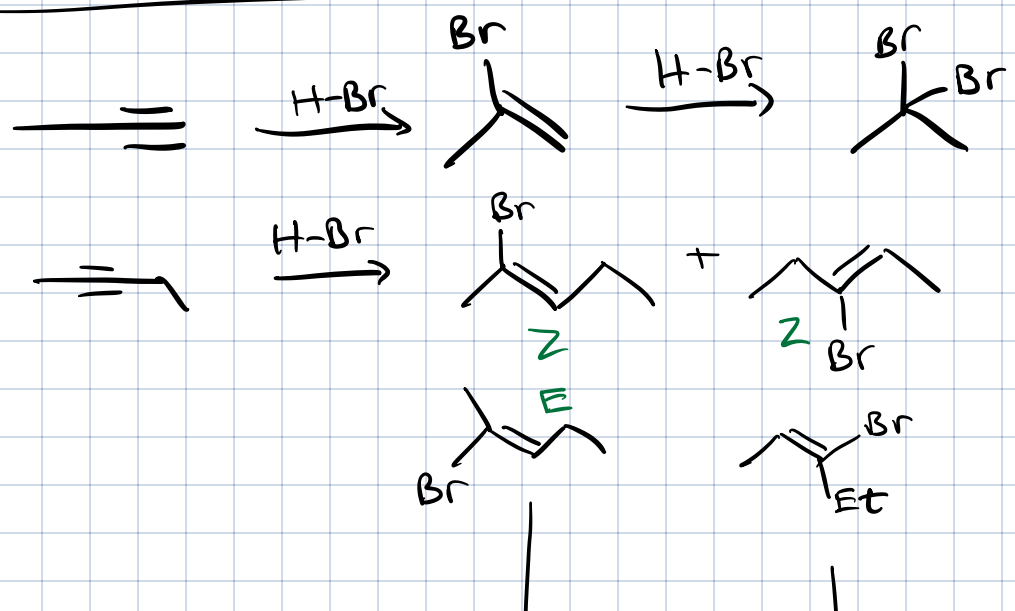


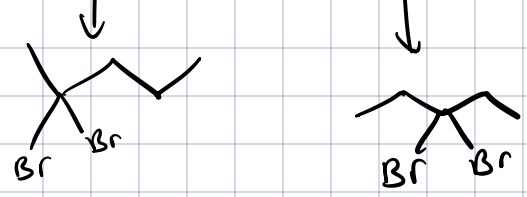
Hydroboration Oxidation

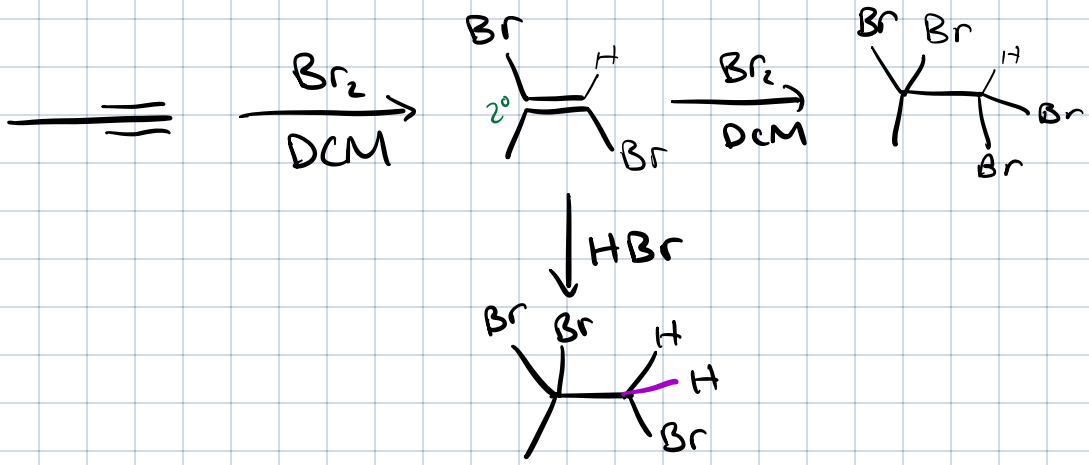


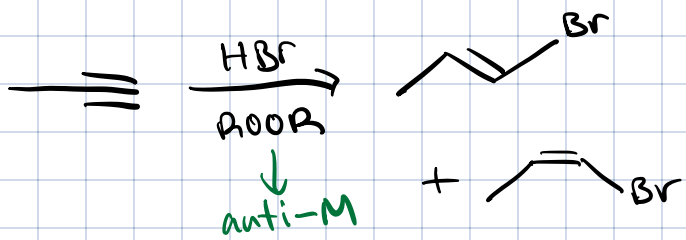



R–C≡C–H
Hg 2+
H2O
δ+
δ+
mercuric enol intermediate
H2O
OH2
Hg +
OH
Hg +
H–O–H
H
enol
OH
Hg+
H2O
OH

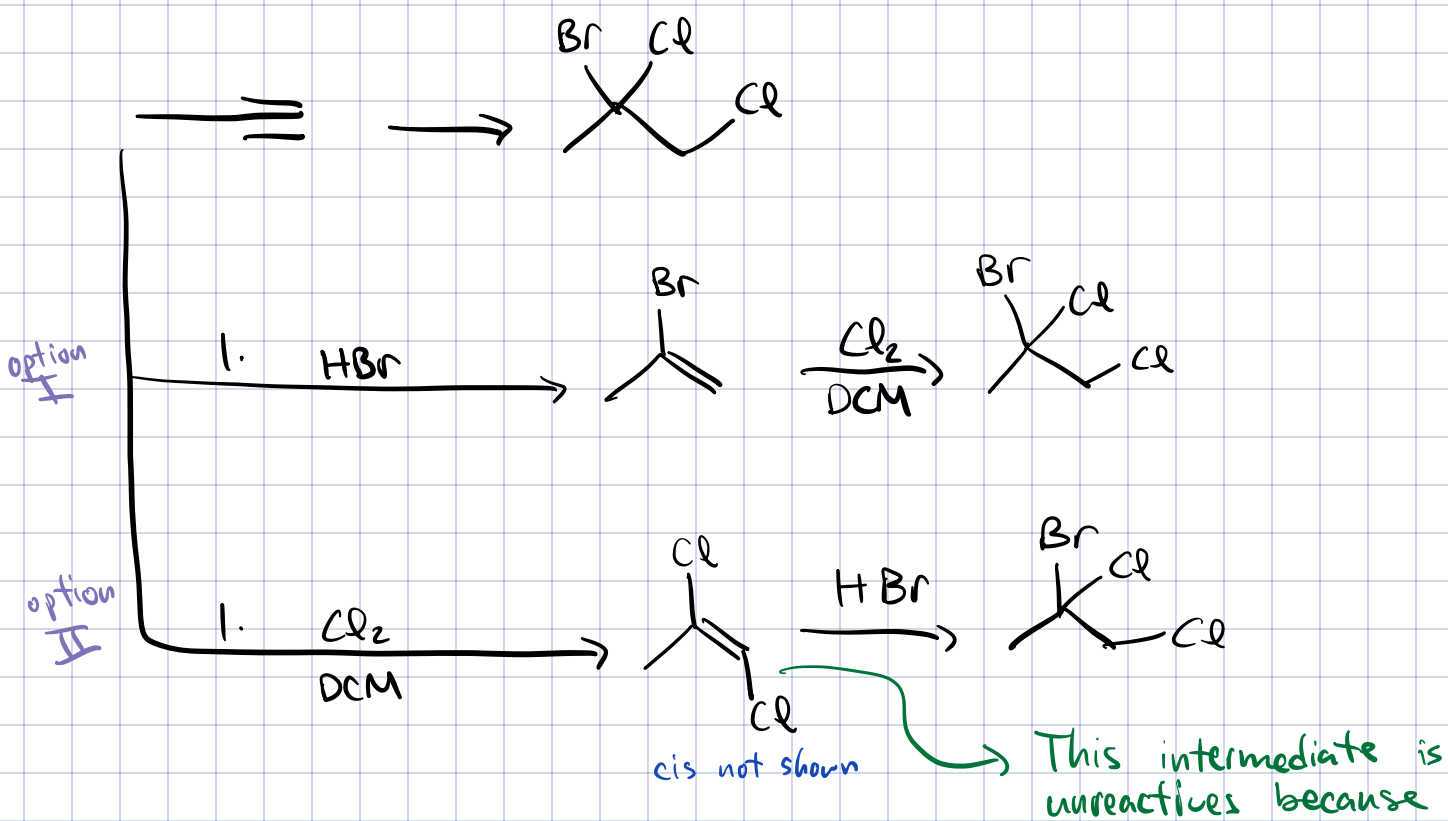

H-Br
Br
H-Br
Br
Br
H-Br
Br
Z
+
Z
Br
E
Br
Br
Et







Beware of intermediate reactivity

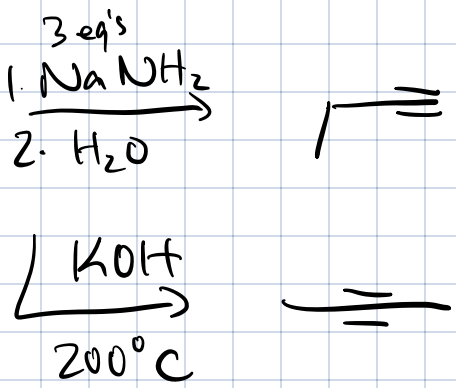


This intermediate is unreactives because

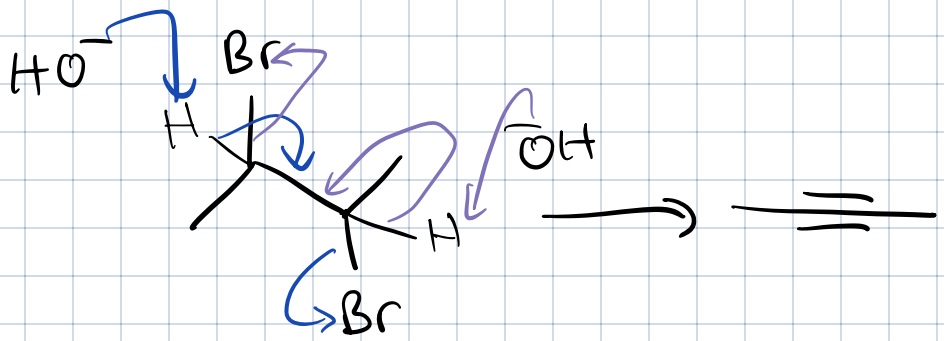
the 2 Cl's withdraw from alkene, making it less Nuph

---

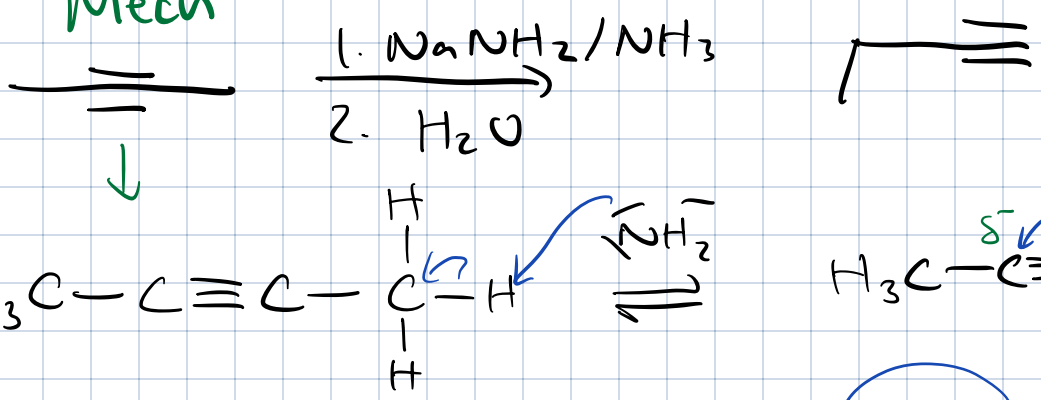
any geminal/vicinal dihalide that is either in 1,1 or 1,2 or 2,2 position

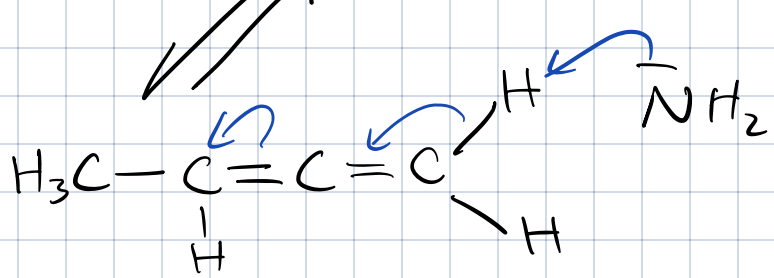


Mech for KOH 200°C

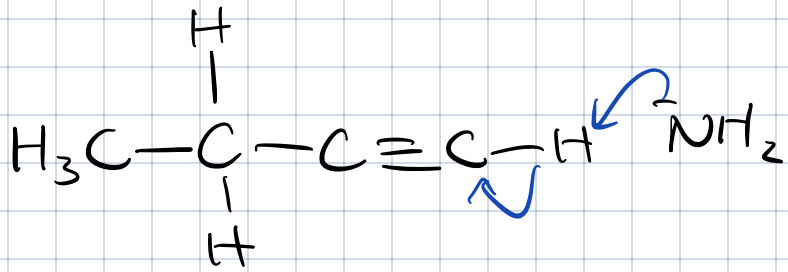


Mech



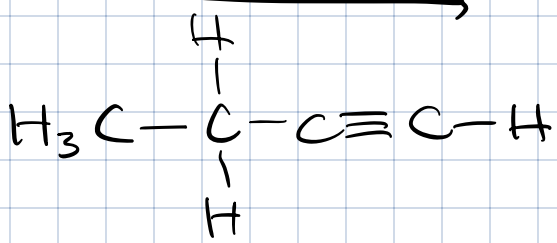


Same thing



workup

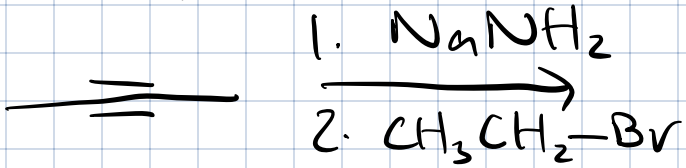
2. H_2O

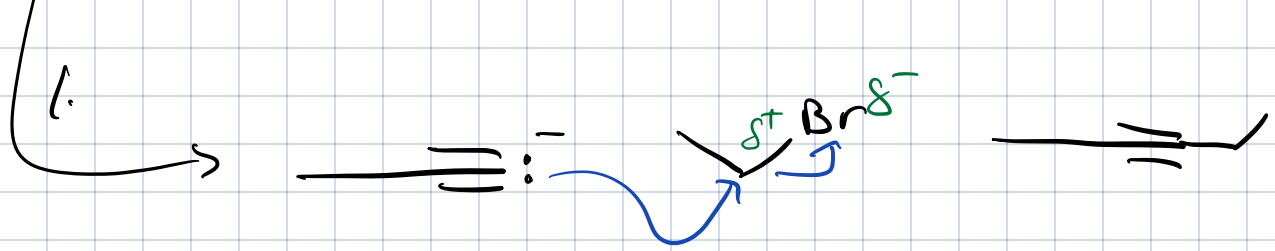


$+ OH^- + NH_3$

KOH needs high temp to push these rxns, but at 200°C, thermodynamic products are favored, so the internal alkyne is formed

Ex:




1.
δ+
Br
δ−